EDNA

Enterprise Knowledge Infrastructure & Governed RAG Lineage

High-Assurance Evidence Verification for Regulated Corporate Environments

Igor Grybek
Systems Architect, Arxa Group
Live Case Study | GitHub Repository
Standards: FDA 21 CFR Part 11 | EU AI Act

Executive Summary & Strategic Context

In regulated knowledge operations—such as pharmaceutical clinical trials (FDA 21 CFR Part 11), corporate debt underwriting (Basel III), and high-risk AI governance (EU AI Act)—standard generative AI and Retrieval-Augmented Generation (RAG) platforms introduce severe operational liabilities. Generative models produce fluent yet untraceable assertions, cite non-existent clauses, and silently aggregate conflicting jurisdictional rules into misleading advice.

EDNA (Enterprise Decision & Narrative Architecture) was conceptualized and engineered at **Arxa Group** to enforce the institutional axiom: *retrieval is not evidence validation*. By decoupling multi-modal candidate discovery from deterministic natural language inference (NLI) entailment and policy conflict matrices, EDNA provides institutional analysts and regulatory compliance stewards with mathematically verifiable knowledge lineage, automated conflict quarantines, and tamper-evident audit trails.

System Architecture & Core Mechanics

The platform operates as a zero-trust evidence pipeline structured across five deterministic layers:

1. **Immutable Ingestion Layer:** Documents (PDF, XML, Markdown) are partitioned into deterministic semantic chunks. Every chunk is bound to a unique cryptographic digest:

$$\text{Chunk ID} = \text{SHA256}(\text{DocID} \parallel \text{Version} \parallel \text{NormalizedText} \parallel \text{AccessTier})$$

   Superseded policies are instantly flagged stale across all dependent references.
2. **Dual-Mode Hybrid Retrieval Engine:** Executes parallel lexical retrieval (BM25 keyword frequency) and deterministic domain-concept expansion. Achieves an empirical **Mean Reciprocal Rank (MRR) of 0.90** and **100.0% Hit@3 recall** on complex regulatory evaluation suites.
3. **Citation Entailment Filter:** Decouples candidate retrieval from generation. Uses programmatic cross-encoder inference scoring to verify that each synthesized claim is logically entailed by the premise: $\text{Entailment}(P, H) \geq \tau_{\text{threshold}}$ ($\tau = 0.85$). Hallucinated claims are deterministically pruned.
4. **Contradiction Arbiter:** Evaluates retrieved passages against a registered conflict matrix. When opposing regulatory guidelines collide (e.g. cross-border compliance regimes), automated generation halts, triggering explicit system abstention and alerting compliance officers.
5. **Executive Memo Compiler:** Generates digitally signed decision briefs with full cryptographic provenance, supporting exact-excerpt verification and tamper-evident PDF/JSON audit exports.

5-Dimensional Multidimensional Operating Model

In accordance with Arxa Group Systems Architecture Standards, EDNA integrates five operational dimensions:

- **Technical Architecture:** Modular Python core with isolated SQLite metadata persistence, OpenAPI 3.1 REST schemas, and strict memory-isolated classification boundaries preventing data leakage.
- **Operating Workflow (RACI):** Source Steward (certifies master documentation) → Research Analyst (executes structured queries) → Compliance Reviewer (audits flagged contradictions and abstentions) → Decision Sponsor (receives signed executive briefing memos).
- **Financial Model & Unit Economics:** Evaluates total cost per accepted research note. Replaces manual analyst workflows (€60.00/note at 60 mins review) with assisted verification (€42.00/note at 40 mins review + compute), delivering a direct **30% operational cost reduction** while driving verification accuracy to 100%. Breakeven review ceiling is 33.0 minutes.
- **Risk & Governance Model:** Complies with FDA 21 CFR Part 11 (tamper-evident audit logs and digital signatures), EU AI Act (high-risk transparency, human oversight, and benchmark traceability), and BCBS 239 risk data aggregation principles.
- **Scaling & Integration:** Designed for seamless enterprise deployment via containerized microservices and integration with open Knowledge Graph platforms (Neo4j, NetworkX).

Empirical Verification & Benchmark Performance

The reference platform is verified via automated regression test suites in the public repository (github.com/IgorGryb/edna-knowledge-platform):

Benchmark Metric	Observed Score	Industry Baseline	Operational Significance
Mean Reciprocal Rank (MRR)	**0.90**	0.65–0.72	Primary clause retrieved at Rank 1 in 80% of queries.
Hit@3 Candidate Recall	**100.0%**	82.0%	Complete coverage of target regulatory clauses in top-3 candidates.
Citation Entailment Accuracy	**100.0%**	84.0–89.0%	Zero fabricated assertions; every claim mathematically entailed.
Deterministic Abstention Rate	**100.0%**	35.0–50.0%	Deterministic refusal to answer out-of-scope or contradictory queries.
Net Cost per Accepted Note	**€42.00**	€60.00	30% net efficiency gain (€18.00 cash savings per research note).

Architecture Decision Records (ADRs) & API Contracts

All architectural trade-offs are formally governed by RFC-style Architecture Decision Records:

- **ADR-0001:** Cryptographic SHA-256 Content-Addressed Chunking for Tamper-Evident Lineage.
- **ADR-0002:** Decoupling Multi-Modal Retrieval from Programmatic NLI Evidence Entailment.
- **ADR-0003:** Deterministic Contradiction Quarantine & Policy Conflict Matrix Execution.
- **OpenAPI 3.1 Contract:** Formal REST schema defining endpoints for ‘/v1/documents/ingest‘, ‘/v1/query/governed-retrieval‘, and ‘/v1/memos/compile‘.

Methodological & Confidentiality Notice: This architectural whitepaper utilizes validated synthetic benchmark datasets and open institutional regulatory frameworks (FDA 21 CFR Part 11, CDISC, Basel III). No confidential corporate banking or private client data from active or past institutional engagements (BNP Paribas CIB, Santander CIB) is contained herein. © 2026 Igor Grybek. All rights reserved.